

Name: _____

Date: _____

PLACE VALUE AND ORDERING INTEGERS, (READING AND WRITING LARGE INTEGERS)

LO: I can read and write integers up to millions using digits and words.

Place value in large numbers

Each digit in a number has a **place value** that depends on its position. From right to left the columns are: ones, tens, hundreds, thousands, ten-thousands, hundred-thousands, millions.

Tip: Group digits in threes from the right (using commas or spaces) to make large numbers easier to read: 3,450,000 is **three million four hundred and fifty thousand**.

Quick examples

●	250,000 = two hundred and fifty thousand	correctly read
●	3,004,500 = three million four thousand five hundred	note the zero place holders
●	47,000 = forty-seven thousand	no hundreds, tens or ones
●	6,300 = sixty-three hundred	should be six thousand three

Key idea

Break the number into groups of three digits: millions | thousands | ones, then read each group.

4, **567**, **890** **4 million, 567 thousand, 890**

Split at the commas, read each group

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - reading a large number

Write 3,250,000 in words.

Split: 3 | 250 | 000

3 million, 250 thousand, 0 ones

In words: three million two hundred and fifty thousand

Three million two hundred and fifty thousand

Group into millions, thousands and ones, then read each section.

Example 2 - writing from words

Write 'seven million forty-five thousand' in digits.

Seven million = 7,000,000

Forty-five thousand = 45,000

Combine: 7,045,000

7,045,000

Write each section as digits, using zeros as place holders.

Example 3 - identifying digit value

What is the value of the digit 6 in 2,600,450?

Position the digit: 2, 6 0 0, 4 5 0

6 is in the hundred-thousands column

Value = $6 \times 100,000 = 600,000$

600,000 (six hundred thousand)

The value of a digit depends on its position in the number.

Example 4 - ordering large numbers

Write these in ascending order: 1,450,000; 1,045,000; 1,500,000; 1,405,000.

Compare millions: all are 1 million

Compare hundred-thousands: 0, 4, 4, 5

Refine: $1,045,000 < 1,405,000 < 1,450,000 < 1,500,000$

1,045,000; 1,405,000; 1,450,000; 1,500,000

When millions are equal, compare the next place value column.

YOUR TURN – PRACTICE (deliberate practice)

1. Reading numbers

Write each number in words.

- a) 450,000
- b) 2,300,000
- c) 5,060,200
- d) 10,500,000

2. Writing numbers

Write each number using digits.

- a) Six hundred thousand
- b) Three million twenty thousand
- c) Nine million five hundred and one
- d) Forty-two million

3. Digit values

State the value of the underlined digit.

- a) The 7 in 7,230,000
- b) The 4 in 1,403,000
- c) The 5 in 6,050,800
- d) The 9 in 3,009,100

4. Ordering large numbers

Arrange each set in ascending order.

- a) 320,000; 302,000; 230,000; 300,200
- b) 5,100,000; 5,010,000; 5,001,000; 5,110,000
- c) 48,000; 408,000; 4,800,000; 480,000
- d) 1,999,000; 2,000,000; 1,990,000; 1,909,000

5. Mixed problems

Answer each question about large numbers.

a) Write the number that is

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) In 3,450,000 the digit 4 represents four hundred thousand ☐

Correct answer: _____

Reason: _____

b) One million is written as 100,000 ☐

Correct answer: _____

Reason: _____

c) 5,060,000 in words is five million and sixty thousand ☐

Correct answer: _____

Reason: _____

d) The number 7,000,300 has no thousands ☐

Correct answer: _____

Reason: _____

e) Grouping digits in threes from the right helps read large numbers ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The population of a city is **2,450,000**. Write this in words. If the population grows by **150,000** each year, write the population after 3 years in both digits and words.

Expression: _____

Simplified: _____

b) A country's national debt is reported as **£45,200,000,000** (45.2 billion). Write this number in words. If the debt decreases by **£1,500,000,000** per year, what will it be after 2 years? Write your answer in digits.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

